



RoadSense

SUPERVISOR
ARSALAN-UL-HAQ

ABDUL AHAD-59882
UMER BIN IBRAR-61072

Outline



- **Introduction**
 - **Problem Statement**
 - **Objectives**
- **Literature Review**
- **Methodology**
- **Results**
- **Project Workload**
- **Gantt Chart**

Introduction

The Problem:

- 1.19 million road deaths annually (WHO, 2021)
- Lack of real-time hazard awareness
- Preventable accidents due to poor road conditions

Our Solution: Transform smartphones into intelligent road monitoring devices:

- **Sensor-driven detection** - Accelerometer + Gyroscope for hazard identification
- **AI-powered prediction** - Hybrid ML model for accident risk assessment
- **Environmental awareness** - Real-time pollution tracking
- **Community intelligence** - Crowdsourced safety data

Impact: Proactive safety through passive monitoring - no additional hardware required

Problem Statement

Critical Gaps in Current Road Safety Systems:

1. Reactive, Not Proactive

- Users only aware of hazards *after* encountering them
- No advance warning system for road anomalies

2. Fragmented Information

- Separate apps for navigation, traffic, pollution
- No unified safety intelligence platform

Problem Statement

Critical Gaps in Current Road Safety Systems:

3. Limited Hazard Detection

- Expensive infrastructure required (sensors, cameras)
- Poor coverage in developing regions

3. No real-time community-based reporting

- Accident Prevention Gap
- Existing systems alert *during* accidents (airbags, ABS)
- Missing: **predictive warnings before dangerous situations**

Objectives



Primary Goal: Develop a cross-platform mobile app that enhances road safety through intelligent monitoring and predictive analytics

Specific Objectives:

1. Real-Time Hazard Detection

- Detect potholes, bumps, and road damage using smartphone sensors
- Achieve $\geq 85\%$ detection accuracy

2. Accident Risk Prediction

- Implement machine learning prediction system
- Alert drivers before dangerous situations occur

Objectives



3. Environmental Awareness

- Integrate real-time pollution and air quality data
- Enable health-conscious route planning

3. Unified Safety Platform

- Combine navigation, hazard alerts, and environmental monitoring
- Single app solution for comprehensive road intelligence

3. Community-Powered Intelligence

- Build crowdsourced hazard database
- Leverage collective data to improve safety for all users

Literature Review

Research Landscape

Sensor-Based Road Monitoring

- Accelerometer/gyroscope for vibration analysis
- 50-100 Hz optimal sampling rate
- Z-axis primary, tri-axial improves accuracy 10-20%

Detection Approaches Evolution

- Threshold-based: Simple, 74-78% accuracy
- Machine Learning: SVM, RF, CNN → 85-98% accuracy
- Deep Learning: LSTM, ETLNet → 95-99% F1-score

Literature Review

Key Methodologies

- Signal preprocessing: High-pass filters, envelope demodulation
- Feature extraction: Time-domain, wavelet (SWT), spectral
- Virtual reorientation: Euler angles for orientation independence

Datasets & Benchmarks

- Pothole Lab Dataset (standard reference)
- US-Accidents (2.25M records)
- RDD2020 (road damage classification)

Literature Review

Deployment Strategies

- TensorFlow Lite for mobile optimization
- Model quantization (32-bit → 8-bit)

Research gaps + Our Contributions

Gap Area	Current State	Limitations
Detection Scope	Road surface only	No accident prediction
Processing	Cloud or offline	Limited real-time capability
Approach	Pure ML or pure rules	Lack of hybrid intelligence
Integration	Single-purpose apps	No environmental awareness
Platform	Native per OS	Maintenance overhead

Literature Review

Our Contributions

- ✓ **Hybrid Architecture:** Rule-based + ML for reliability + adaptability
- ✓ **Real-Time On-Device:** <200ms processing latency
- ✓ **Dual Prediction:** Road anomalies + accident risk assessment
- ✓ **Multi-Sensor Fusion:** Accelerometer + Gyroscope + GPS
- ✓ **Environmental Integration:** Pollution data via Google Maps API
- ✓ **Cross-Platform:** Flutter for Android + iOS from single codebase

Methodology

Data Acquisition

- Smartphone sensors: Accelerometer + Gyroscope (50 Hz)
- GPS positioning with timestamp synchronization
- Real-time location tracking (100ms update intervals)

Signal Processing

- Exponential smoothing filter ($\alpha=0.3$)
- Feature extraction: 238-sample window (~ 4.76 s)
- Multi-modal trigger: acceleration + rotation variance
- Speed-adaptive thresholds (5 tiers: 2-150 km/h)

Methodology

Classification Algorithm

- Weighted scoring system (5 features):
 - Peak sharpness, gyro variance, vertical dominance
 - Impact duration, frequency analysis (FFT)
- Pothole vs Speed Bump discrimination
- Confidence calculation ($\geq 60\%$ minimum threshold)

Pollution Monitoring

- Google Air Quality API integration
- Heat map visualization with concentric zones

Methodology

Navigation System

- Google Maps Directions API with route alternatives
- Turn-by-turn guidance with lane assistance
- Off-track detection (30m threshold)
- Automatic rerouting (10s cooldown)

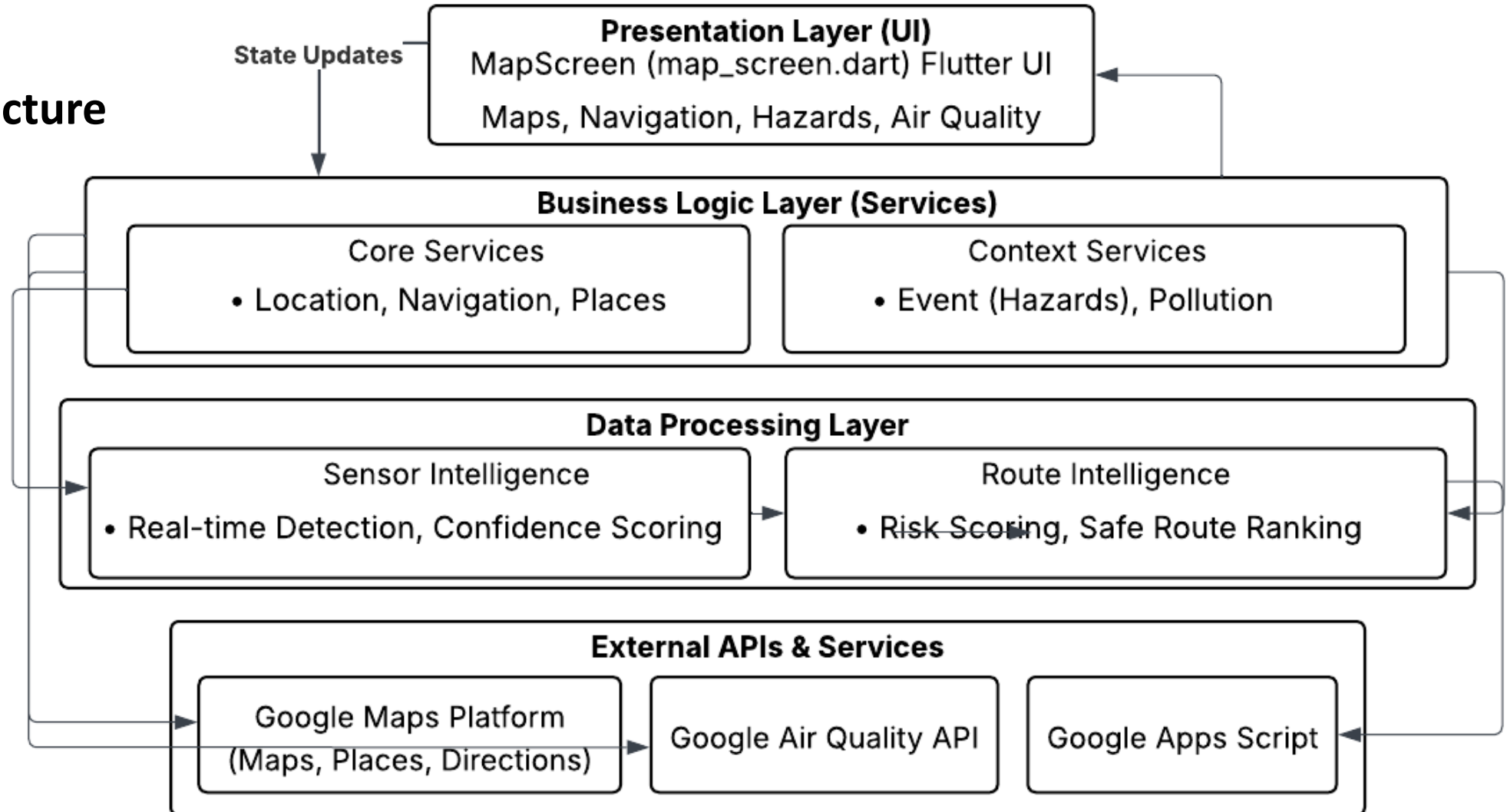
Route Risk Analysis

- Hazard proximity detection (50m threshold)
- Multi-criteria scoring: hazards + pollution + distance
- Safest route recommendation

Methodology

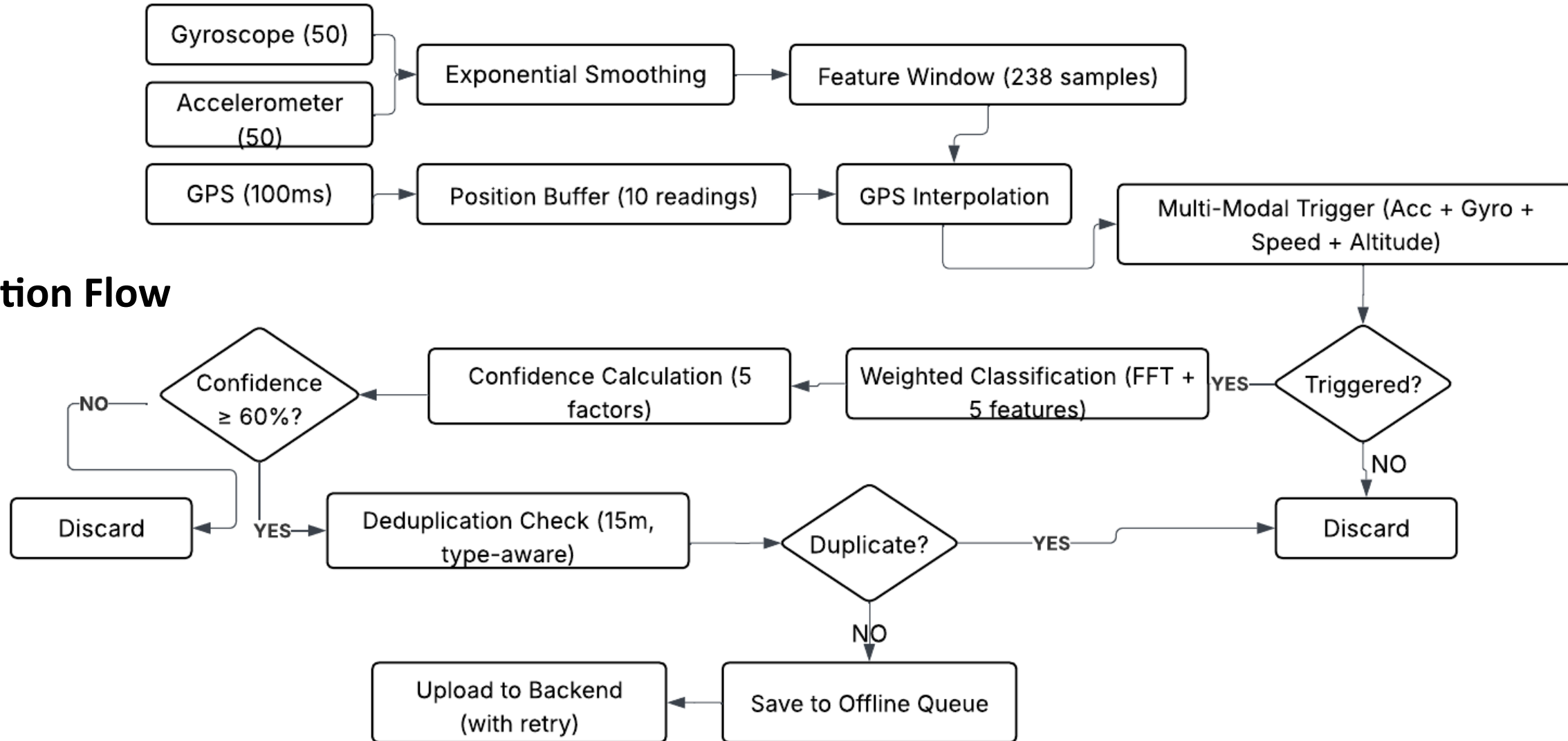


System Architecture



Methodology

Event Detection Flow



Results

Data Quality Filtering

Raw detections: 61

Recent events (30 min): 19

Consensus-validated events: 8

False positives filtered: 86.9%

Detection Accuracy

Average initial confidence: 87.4%

Post-consensus confidence: 100%

Event distribution:

- Potholes — 62.5%
- Speed bumps — 37.5%

Aggressive filtering ensures only high-confidence events persist.

All validated events confirmed by ≥ 2 independent devices.

Results

Real-Time System Performance

GPS updates: 100 ms

Backend response: < 1 s

Auto-refresh cycle: 30 s

Platforms tested: Android

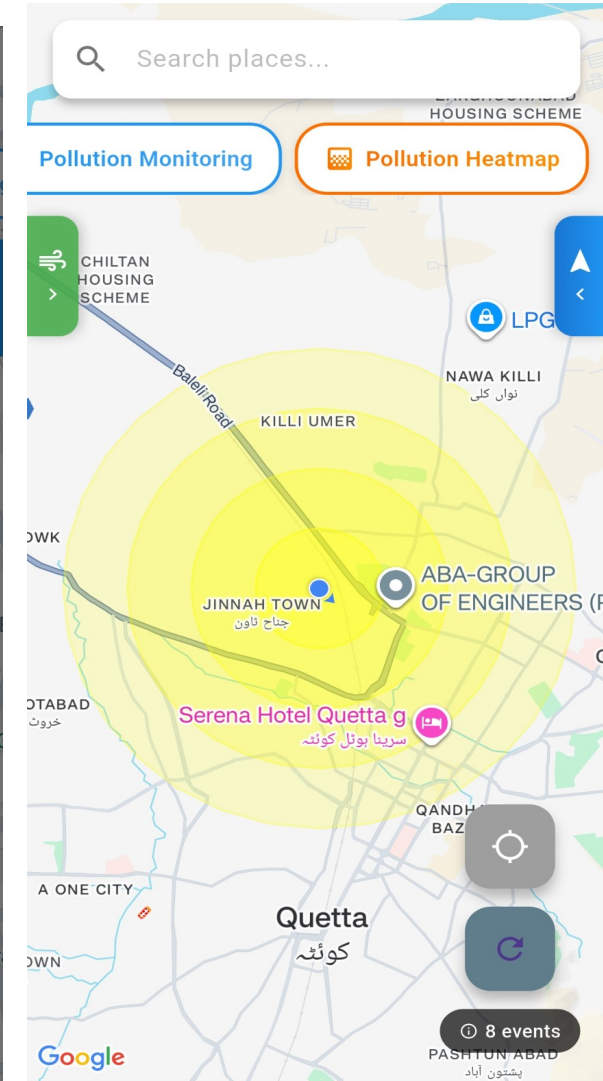
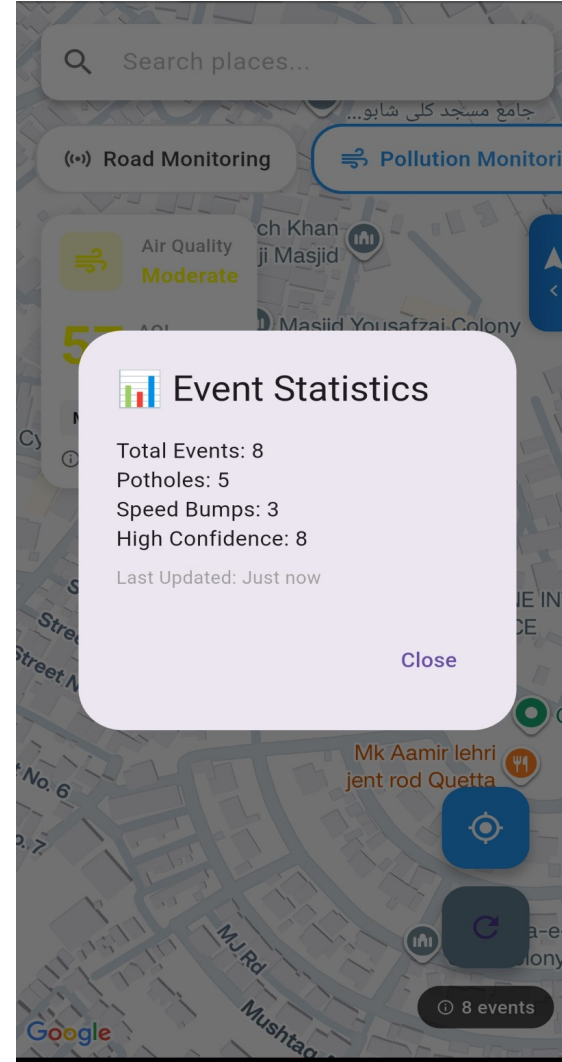
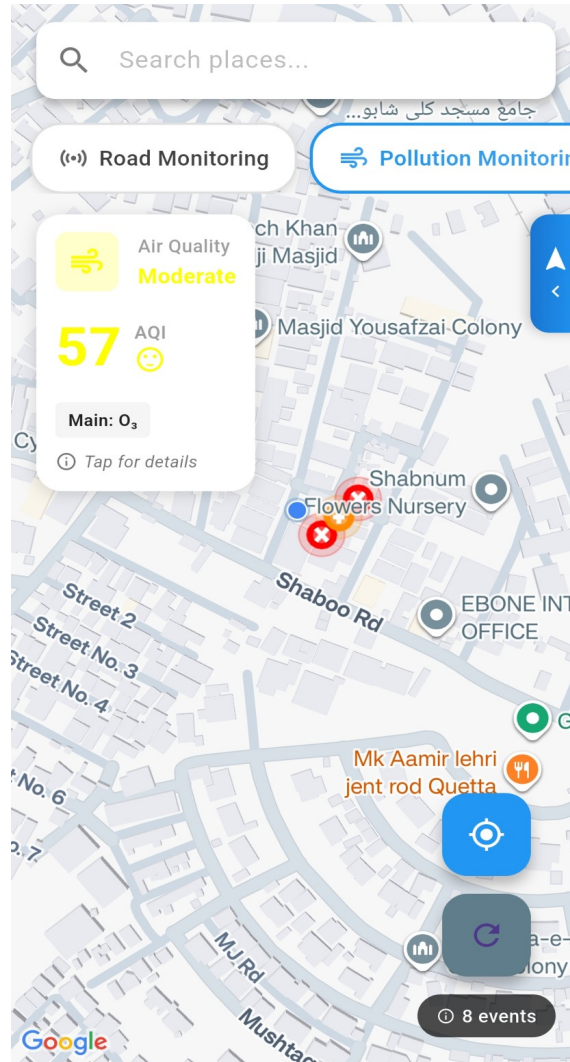
Real-world deployment: Quetta, Pakistan

Key Outcome

Reliable, real-time, crowd-validated road hazard detection using mobile sensors

Results

System in Operation



Project Workload

Abdul Ahad — Core System & Intelligence

- System architecture & service-layer design
- Sensor data processing (accelerometer, gyroscope)
- Road anomaly detection logic
- Consensus validation & noise filtering
- GPS tracking and real-time location handling
- Backend integration (event upload & sync)
- Performance optimization & result analysis

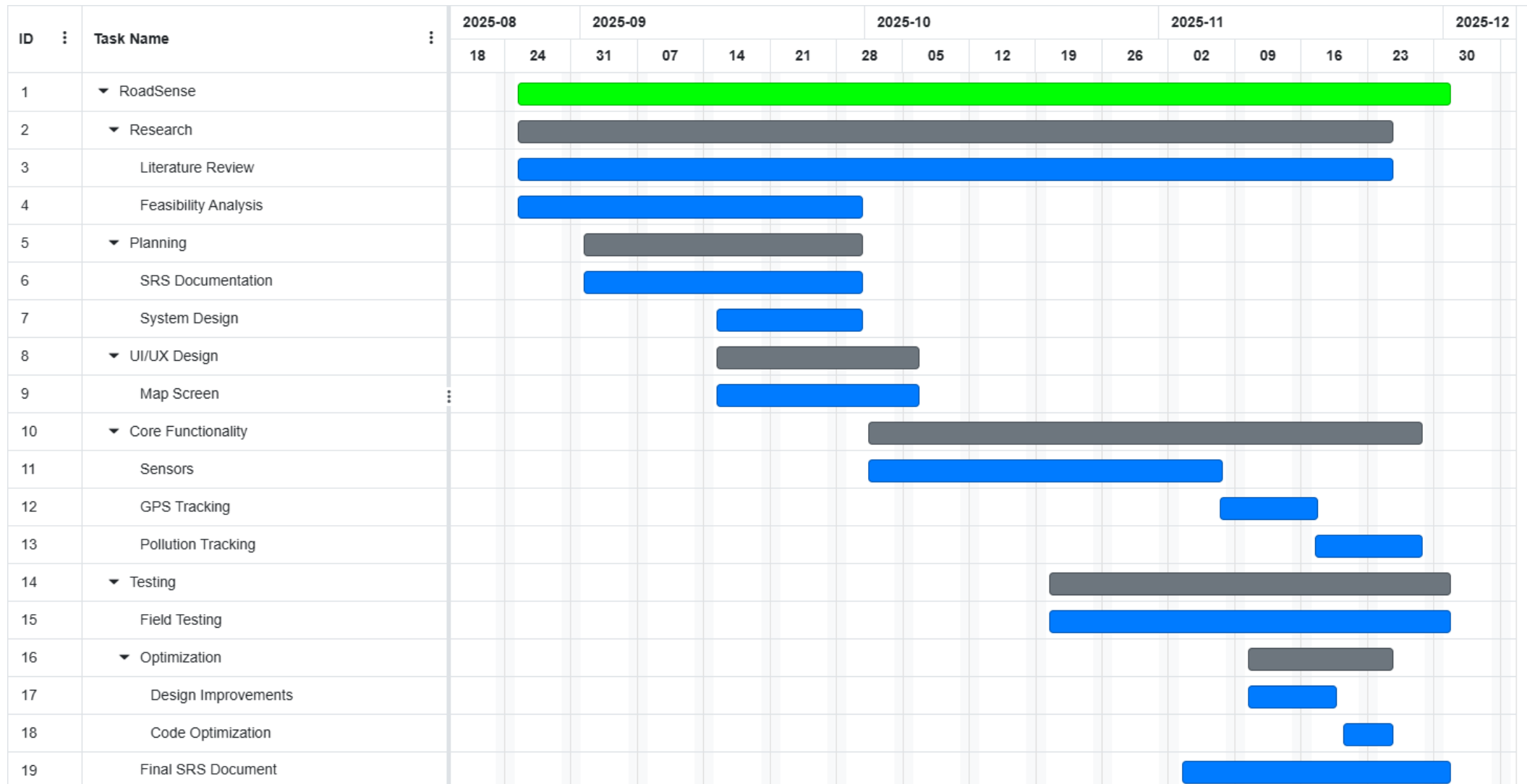
Focus: core logic, data reliability, system performance

Umer Bin Ibrar — UI, Maps & User Interaction

- Flutter UI development & screen flows
- Google Maps integration & visualization
- Hazard markers & pollution overlays
- Navigation interface & user alerts
- Cross-platform consistency (Android & iOS)
- UI/UX improvements and usability testing

Focus: user experience, visualization, interaction design

Gantt Chart



References

- [1] M. Zareei et al., "Machine Learning Model for Road Anomaly Detection Using Smartphone Accelerometer Data," *IEEE Access*, 2025. (98% accuracy, CNN-based, adaptive thresholding)
- [2] P. Mohan et al., "TrafficSense: Rich monitoring of road and traffic conditions using mobile smartphones," Microsoft Research, 2008. (Foundational work, virtual reorientation)
- [3] F. Seraj et al., "RoADS: A road pavement monitoring system for anomaly detection," *Big Data Analytics*, 2014. (Envelope demodulation, 90% accuracy)
- [4] A. Allouch et al., "RoadSense: Smartphone application to estimate road conditions," *IEEE Sensors J.*, 2017. (98.6% accuracy, C4.5 Decision Tree)
- [5] O. Mazari Abdessameud & W. Cherifi, "Federated road surface anomaly detection," *DATA Conf.*, 2024. (LSTM + federated learning)

Thank You!